Part III: Applications, Trading & Risk Management

Volume Risk, Flexibilities, and Real Options in Engineering Assets

Commodity Markets and Models

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Lecture Plan: Part III

Course Outline

1. **3.1 Risk beyond price:** Volume uncertainty, production risk, and the "Natural Hedge."
2. **3.2 Valuing flexibility:** Swing contracts, Take-or-Pay, and Storage as a calendar spread option.
3. **3.3 Physical assets as real options:** Valuation of Power Plants (Spark-spread), Refineries (Crack-spread), and Transportation (Locational-spread).

3.1.1 The Multi-Dimensional Nature of Risk

Definition: Volume Risk

The uncertainty regarding the quantity Q to be produced, delivered, or consumed at a future date T. This risk is often exogenous to the market price.

Engineering Drivers of Volume Uncertainty:

- **Supply Side:** Weather events (droughts, floods, frost), bush fires, or unplanned technical outages in mines/refineries.
- **Demand Side:** Temperature sensitivity in power and gas (Heating/Cooling Degree Days).

Key Insight: Incomplete Markets

Many spot positions cannot be hedged by futures or forward trading. Consider a power retailer delivering variable load: even with perfect demand forecasting, no traded futures contract offsets hourly spot price exposure.

3.1.2 Commodity Swaps: Foundation for Flexibility

Definition: Commodity swaps are OTC derivatives where two parties exchange:

- **Fixed leg:** Fixed price G (guaranteed price) for a defined quantity Q
- **Floating leg:** Spot price $S(t_j)$ times quantity Q at each settlement date

Swap Value Formula

At date t, the value of a long swap position is:

$$V_{\text{swap}}(t) = \sum_{j=1}^{q} e^{-r(t_j - t)} Q[F^{t_j}(t) - G]$$

where $F^{t_j}(t)$ is the forward price for delivery at t_j.

Key Features

- Financially settled (no physical delivery)
- Customized terms (OTC)
- Foundation for more complex structures like swaptions
- Requires reliable index (e.g., Platts) for floating leg determination

3.1.3 The Revenue Equation and Correlation

For a producer, the gross revenue R is a product of two stochastic variables:

$$R_T = P_T \times Q_T \tag{3.1}$$

Notations:

- P_T: Spot price at time T.
- Q_T: Realized volume at time T.

The Natural Hedge Concept

In commodities like agriculture or power, price and volume are often negatively correlated ($\rho_{P,Q} < 0$).

- **Low Yield ($Q \downarrow$)** $\implies$ Scarcity $\implies$ **High Price ($P \uparrow$)**.
- This correlation reduces the variance of R, but makes pure price hedging (futures) dangerous if Q is not known.

3.1.4 Full Requirement Contracts: Case Study

Power Retailer's Problem

Must serve random client demand $d_1, \ldots, d_N$ at fixed price K, with penalty P for unmet demand.

Cashflow decomposition:

$$\begin{aligned} CF_i &= d_i(K - S_i) + (S_i - P)s_i \\ &= \underbrace{d_i(K - S_i)}_{\text{Swap with random volume}} + \underbrace{(S_i - P)s_i}_{\text{Call option part}} \end{aligned}$$

Hedging Challenge

- Swap part depends on both prices *and* demand
- Demand negatively correlated with temperature
- Weather derivatives market often illiquid
- Higher correlation $\rightarrow$ higher retailer exposure

Crop Uncertainty

Agricultural producers face dual uncertainty:

- Selling spot price (hedgeable)
- Crop volume (affected by weather)

Hedge via crop insurance or weather derivatives.

3.1.5 Commodity Futures Markets & Hedging

Why Use Futures Instead of Physical?

- Avoid storage, insurance, perishability issues (coffee, natural gas, electricity)
- Only need margin deposit (small % of contract value)
- No physical delivery concerns if position closed before maturity

Commodity Futures Characteristics

- Traded on exchanges (CME, CBOT, NYFE) with clearing house integrity
- Initial margin typically 5-15% of contract value
- Can be settled financially or by offsetting position
- Roll strategy critical for longer investment horizons

Example: Coffee Futures

A Dutch fund buying 1M coffee bags from Vietnam faces storage issues: space, humidity, perishability. Futures avoid these while providing price exposure.

3.1.6 Agricultural Production Risk & Insurance

Production Risk Factors

- Weather events: droughts, floods, frost, heat waves
- Disease outbreaks
- Bush fires
- Soil erosion (10-100x faster than replacement)

Insurance Instruments

- **CAT coverage**: Catastrophic event insurance
- **Crop revenue insurance**: Revenue below predefined level
- **Crop yield insurance**: Yield below predefined level
- Weather derivatives (temperature, rainfall)

Global Land Pressure

- 1950: 0.5 ha arable land/person $\rightarrow$ 2020: 0.2 ha/person
- Food production must increase 70% by 2050 vs 2005-2007
- US losing arable soil 10x faster than replacement

3.2.1 Commodity Swaptions: Options on Swaps

Definition: A swaption is an option to enter into a swap at a future date T at a pre-agreed strike price G.

Payoff at Maturity

$$C_{\text{swap}}(T) = \max(0, V_{\text{swap}}(T))$$

where $V_{\text{swap}}(T)$ is the value of the underlying swap at time T.

Key Features

- Provides optionality on future swap entry
- Popular in interest rate markets, now in commodities
- Requires assumption about swap value dynamics

Pricing Approaches

- Arithmetic Brownian motion for swap value
- Forward curve dynamics using formula (12.2)
- Volatility of swap price is key parameter

3.2.2 From Swaps to Swing Contracts

Standard Swap Limitation

Fixed quantity Q at fixed price G over $T_1, \ldots, T_N$:

$$SW = \frac{\sum_{i=1}^{N} B(t, T_i) F(t, T_i)}{\sum_{i=1}^{N} B(t, T_i)}$$

No volume adjustments allowed.

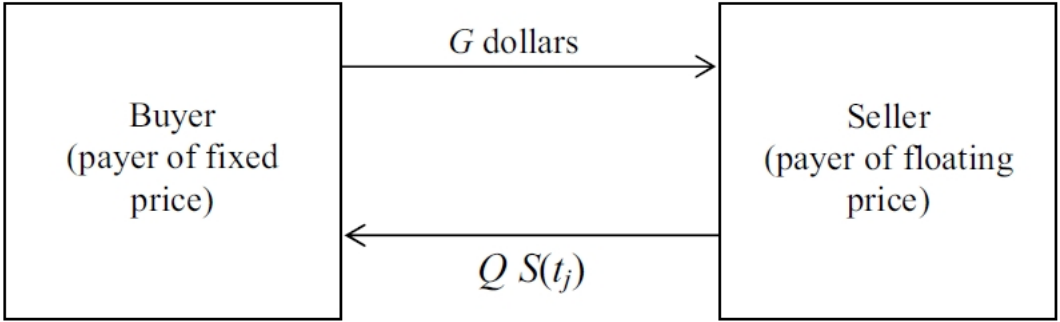
Swing Contract Solution

Flexible purchase agreements where buyer has quantity flexibility at settlement dates. Purchase price fixed or indexed; quantities subject to bounds.

Real Option Analogy

A copper or oil concession with known reserves and fixed exploitation horizon can be assimilated to a swing contract with fixed cumulative "purchase".

The buyer of the swap pays a fixed amount of G dollars – G is the guaranteed price of the swap – corresponding to a well-defined quantity Q of the commodity underlying the swap contract



3.2.3 Mathematical Definition of Swing Options

A swing contract allows the holder to choose daily quantity q_t subject to constraints:

Mathematical Constraints

$$m \leq q_t \leq M \qquad \text{(Daily limits)} \tag{3.2}$$

$$A \leq \sum_{t=1}^{N} q_t \leq B \qquad \text{(Cumulative limits)} \tag{3.3}$$

Special Cases:

- $m = 0$: No minimum daily withdrawal
- $A = B$: Fixed total take across period (Take-or-Pay)
- M: Maximum daily withdrawal (seller's capacity constraint)

Optimal Strategy Complexity

The optimal exercise strategy is NOT simply "all or nothing" when spot prices are stochastic. Buyers may recall lower $q(t)$ today to reserve capacity for higher $q(t+1)$ tomorrow when prices spike.

3.2.4 American Options: Foundation for Swing Valuation

Definition: Options that can be exercised at any time τ during $[0, T]$.

Valuation Challenges

- Optimal stopping time problem
- No closed-form solution in general
- Path-dependent with buyer strategy

Tree Method (Binomial)

Backward recursive approach:

$$V(t_{N-1}, S) = \max\left(S - k, \frac{\pi V_u + (1-\pi)V_d}{1+r}\right)$$

Divides (t, S) plane into exercise and continuation regions.

Bounds

- Lower bound: European option or intrinsic value
- Upper bound: Spot price S (arbitrage)
- Bermudan: Exercise on finite dates (simpler)

3.2.5 Storage as a Calendar Spread Option

Storage facilities (e.g., salt caverns for gas) allow for arbitrage across time.

$$V_{storage} = \max(S_{t_{out}} - S_{t_{in}} - C_{cost}, 0) \tag{3.4}$$

Notations:

- $S_{t_{out}}$: Spot price at withdrawal time
- $S_{t_{in}}$: Spot price at injection time
- C_{cost}: Injection/withdrawal fees + storage costs

Seasonal Arbitrage Opportunities

- **Gas**: Winter/summer spreads (heating demand)
- **Power**: Weekly spreads (weekdays/weekends)
- **Contango markets**: Lock value by longing short-term futures and shorting long-term futures

3.2.6 Gas Storage: Types and Optionality

Types of Underground Storage:

- **Depleted fields:** Low turnover, seasonal summer/winter spreads
- **Aquifers:** Water-bearing rock with impermeable cap
- **Salt caverns:** High deliverability, multiple turnovers possible

Storage Optionality Components

- Option on seasonal spread (price as underlying)
- Option to switch injection/withdrawal modes
- Option to time hedging decisions
- Option to deliver to multiple pipelines

Operational Constraints

- Injection/withdrawal rates depend on storage level
- Contractual obligations (Take-or-Pay)
- Location value (proximity to pipelines/hubs)

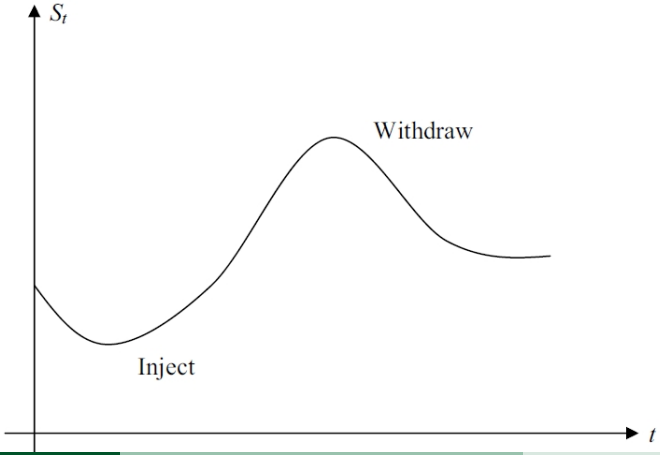
3.2.7 Ways to Gain Commodity Structured Products & Indices

- Physical commodities (storage/insurance costs)
- Futures contracts (most common)
- Options on futures
- Commodity index investing (GSCI, DJ-UBS, CRB)
- Commodity-linked bonds
- Structured products (principal-protected notes)
- ETFs/ETNs
- Mining/oil company shares
- Physical assets (transformation spread)
- Land acquisition

Principal-Protected Structured Product

- Popular in equity/commodity markets
- Value: $S(t) + P(t) =$ commodity + put option
- Guarantees minimum return (floor) while maintaining upside
- Example: 95% principal protected note on Goldman Sachs Agriculture Index

Withdrawal and injection rates constrain the possible values. The seasonality of gas prices and changes in the shape over time of the forward curve are crucial to the value of a storage facility



3.2.8 Exchange Options: Margrabe's Formula

Definition: Option to exchange one asset S_2 for another S_1.

Payoff at Maturity

$$C(T) = \max(0, S_1(T) - S_2(T))$$

Margrabe's Formula (1978)

$$C(t) = S_1(t)e^{-y_1(T-t)}N(d_1) - S_2(t)e^{-y_2(T-t)}N(d_2)$$

where:

$$d_1 = \frac{\ln\left(\frac{S_1}{S_2}e^{-(y_1-y_2)(T-t)}\right) + \frac{1}{2}\Sigma^2(T-t)}{\Sigma\sqrt{T-t}}$$

$$d_2 = d_1 - \Sigma\sqrt{T-t}$$

$$\Sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$$

Key Insight

- No interest rate in formula
- Only relative volatility Σ matters
- Vega can be negative if $\sigma_1 - \rho\sigma_2 < 0$
- Common in energy (gas vs electricity)

3.2.9 Commodity Spread Options

General Form:

$$C(T) = \max(0, Q_1 S_1(T) - Q_2 S_2(T) - k)$$

where $k > 0$ is the strike.

Major Types in Energy

- **Crack Spread:** Refined products vs crude oil
- **Spark Spread:** Electricity vs fuel (gas, coal, oil)
- **Dark Spread:** Electricity vs coal (with emissions)
- **Calendar Spread:** $F^{T_1}(T) - F^{T_2}(T)$ with $T_1 < T_2$

Valuation Approaches

- Geometric Brownian motion for each leg
- Mean-reverting processes
- Monte Carlo simulation
- Cholesky decomposition for correlation

Monte Carlo Pricing

$$C^{MC}(0) = e^{-rT} \frac{1}{N} \sum_{i=1}^{N} \max(0, Q_1 S_1^i(T) - Q_2 S_2^i(T) - k)$$

3.2.10 Numerical Valuation: Monte Carlo & LSMC

Because swing and storage options are path-dependent, we use numerical methods:

Least-Squares Monte Carlo (LSMC)

1. Simulate price paths forward
2. At each exercise date, regress continuation values
3. Compare immediate exercise vs. continuation
4. Choose optimal policy backward

Tree Methods

Best for:

- American/Bermudan exercise
- Low-dimensional problems
- Computing conditional expectations

Discrete Hedging Risk

Under discrete-time hedging, P&L depends on spread between pricing variance and realized variance. Seller earns if realized variance $<$ pricing variance.

Illustration of the Gamma risk under discrete-time hedging

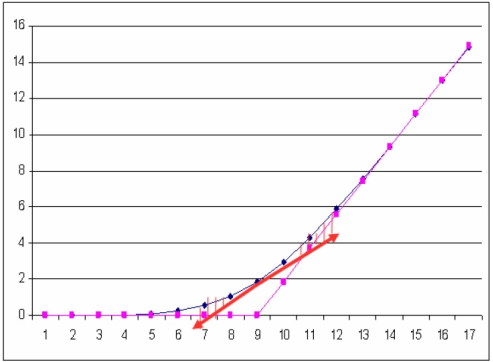


Figure: Delta hedger moves along tangent to option price curve

The delta hedger moves along the tangent to the option price curve. Hence, the seller of the option looses money regardless of the direction of the move : the loss increases with the Gamma of the option, which indicates the local convexity of the curve at the time of the hedge.

3.2.11 Intrinsic vs. Rolling Intrinsic Value

Intrinsic Value

Value that can be locked in *today* using forward/futures contracts:

$$IV(t) = \max_{\text{feasible}} \sum (F_{out} - F_{in} - C)$$

Deterministic optimization using current forward curve.

Extrinsic Value

Additional value from ability to adjust as prices evolve:

$$EV = \text{Total Value} - IV$$

Captures monthly flexibility value.

Rolling Intrinsic Methodology

1. Lock in initial intrinsic value using forwards
2. As prices change, compute new intrinsic value
3. Unwind hedge if: New IV - Unwind Cost > Current position
4. Expected value of future adjustments = extrinsic value

Example: Storage Tank

Initial contango: $F_{T1} = 100$, $F_{T2} = 110$, cost $C = 5 \rightarrow IV = 5$. Curve flips to backwardation at t': $F'_{T2} < F'_{T1} \rightarrow$ new $IV' = 0$. Unwind if $IV' - (F'_{T2} - F'_{T1} - C) > 0$.

3.3.1 The Real Options Paradigm

Concept: Intrinsic vs. Extrinsic Value

The value of a physical asset (plant, mine, ship) is not just the NPV of future cash flows. It includes the **Extrinsic Value** derived from operational flexibility (the option to shut down).

Key Insight

Traditional DCF undervalues assets by ignoring:

- Value of waiting for better information
- Value of operational flexibility
- Asymmetric payoff (limited downside, unlimited upside)

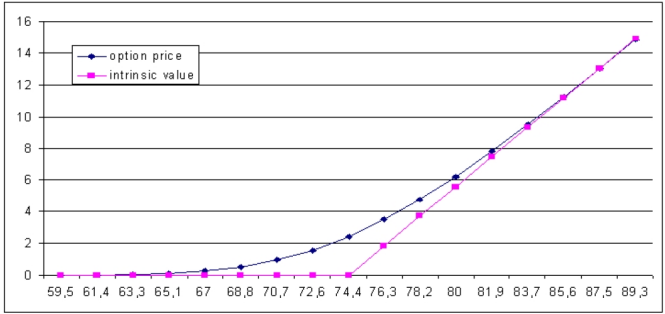
Convexity

Real option value $\geq$ Intrinsic value

$$\frac{\partial^2 V}{\partial S^2} > 0$$

(Convex payoff)

Option price vs intrinsic value



(a) Option price as a function of the initial futures price with constant strike $K = 74.43$ at t_0

Figure: Convexity: The asset value is always $\geq$ Intrinsic Value, especially when volatility is high.

3.3.2 Power Plant Characteristics for Valuation

Technical Parameters

- **Capacity:** MW output (nuclear: 1000+, coal: 500+, gas: 200-800)
- **Heat Rate:** Efficiency (Btu/KWh)
- **Ramp Rate:** MW/minute adjustment speed
- **Min/Max Generation:** Technical limits

Cost Components

- **O&M Costs:** $/unit generation
- **Startup/Shutdown Costs:** Fuel and maintenance
- **Outage Rates:** EFOR = outage hours/total hours
- **Scheduled Maintenance:** 2-4 weeks annually

Outage Risk Impact

Forced outages during price spikes can cause heavy losses when generator has commitments but must buy at high spot prices (ECAR region, June 1998 example).

3.3.3 Power Plants and the Spark Spread

A gas-fired plant converts Natural Gas into Electricity. Its value is a strip of **Spark Spread Options**:

$$\text{Payoff}(T) = \max(S_e(T) - H_{eff} \cdot S_g(T) - K, 0) \tag{3.5}$$

Notations:

- $S_e(T)$: Electricity price (\$/MWh)
- $S_g(T)$: Natural gas price (\$/MMBtu)
- H_{eff}: **Heat Rate** (Efficiency, MMBtu/MWh)
- K: Variable operating and maintenance costs

Physical Constraints

- Minimum up/down times
- Ramp rate limits
- Start-up/shut-down costs
- Maintenance requirements

Valuation Complexity

Not a strip of independent options! Intertemporal constraints create path dependence requiring:

- Dynamic programming
- Stochastic control
- Multi-factor models

3.3.4 Spark Spread Options: Definition & Importance

Precise Definition

A spark spread option is a European call on the difference between electricity price and fuel (natural gas) price:

$$\max\left(S_e(T) - H_{eff}\, S_g(T) - K, 0\right)$$

where:

- $S_e(T), S_g(T)$: Futures prices for electricity and gas
- H_{eff}: Heat rate (efficiency factor, Btu/KWh)
- K: Strike price

Units: Electricity in \$/MWh, Gas in \$/MBtu

Economic Importance

- **Hedging:** Risk mitigation for power plant operators
- **Valuation:** "Paper plants" - represent value of converting gas to electricity
- **Real Options:** Alternative to NPV for new plant evaluation (Geman 2005a)
- Spark spread options are "**by far the most important correlation product in the energy context**" - they approximate the cost of converting gas into electricity.

3.3.5 Spark Spread Pricing Challenges

Analytical Limitations

- **No closed form** when $K \neq 0$ (even in Black-Scholes)
- For $K = 0$: Exchange option (Margrabe formula)
- For $K \neq 0$: Monte Carlo or approximations (Kirk, Bachelier, Carmona-Durrleman bounds)

Modeling Challenges

- **Commodity price features:** Skewness, kurtosis, deviations from normality
- **Lévy models:** Include diffusive Brownian motion + pure jump processes
- **Time-changed processes:** Represent information/trade flow economically

Joint Dynamics Problem

- **Bivariate processes:** Scarce literature on different time changes for different assets
- **Existing approaches:**
  - Model spread as single Lévy process (Benth and Benth 2006) - loses consistency
- **Need:** Well-defined multivariate Lévy process consistent with both margins

3.3.6 Time-Changed Processes for Commodities

Business Time vs. Calendar Time

- **Clark (1973):** "On days when no new information is available, trading is slow... On days when new information violates old expectations, trading is brisk"
- **Business time:** Runs according to market activity, information flow, order arrival
- **Monroe (1978):** Any semimartingale is a time-changed Brownian motion

Log return representation:

$$Y(t) = \mu G(t) + \sigma B(G(t))$$

- $B(G(t))$: Brownian motion evaluated at time $G(t)$
- $G(t)$: Pure jump time-changing process (subordinator)
- μ, σ: Constant parameters

Why Pure Jump?

- Geman et al. (2001): Time changes are "purely discontinuous if not locally deterministic"
- Pure jump time changes generate pure jump price processes
- Particularly useful for commodity prices with frictions (delivery needs, imbalances)

3.3.7 Application: PJM Electricity & NYMEX Natural Gas

Empirical Study

- **Data:** 79 PJM electricity options, 49 NYMEX natural gas options (5 Nov 2007)
- **Models tested:** BS, VG, NIG, Meixner, CGMY, GH
- **Calibration:** Minimize pricing errors using Fourier space-time stepping

Results

- **BS severely underperforms:** Errors twice as large as pure jump models
- **Best models:** GH and VG for both electricity and gas
- **Moments:** Negative skewness, pronounced kurtosis (non-Gaussian)

Optimal Parameters

- VG (PJM): $\sigma = 0.4352$, $\mu = -0.04307$, $\alpha = 2.0199$
- VG (Nat Gas): $\sigma = 0.5048$, $\mu = -0.1254$, $\alpha = 1.3668$

Key Finding

Pure jump models (VG, GH) provide excellent fit to energy option data, capturing the non-Gaussian features of commodity prices.

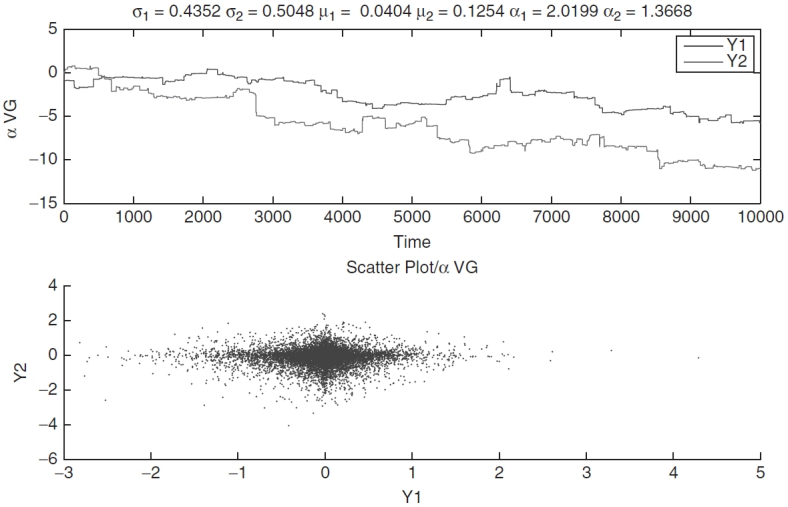


Figure: An example of path and a scatter plot at time 1 of the electricity (Y1) and gas returns (Y2), decorrelated case. The path has been obtained with 10 000 time steps: the number of time steps appears on the horizontal axis.

3.3.8 Refineries and the Crack Spread

Refineries process Crude Oil into Products (Gasoline, Diesel, Heating Oil):

$$\text{Crack Spread} = \sum w_j S^j_{prod} - S_{crude} - C_{refining} \tag{3.6}$$

Industry Benchmarks

- **3:2:1 Crack**: 3 crude $\rightarrow$ 2 gasoline + 1 heating oil
- **5:3:2**: 5 crude $\rightarrow$ 3 gasoline + 2 diesel
- Product weights based on typical yield patterns

Refinery Optionality

- Option to vary throughput (crude runs)
- Option to change product slate
- Option to switch crude types
- Option to store vs. immediate sale

Transformation Plant Generalization

Any flexible plant converting product A to B can be assimilated to an option on spread between prices of B and A. Complexity arises from intertemporal constraints (minimal shutting/starting times, minimal operation time).

3.3.9 DCF vs Real Options: Power Plant Valuation

DCF Approach

$$V(0) = \sum_{j=1}^{n} \frac{E(\tilde{\phi}_j)}{(1+r^a)^j}$$

where:

$$\tilde{\phi}_j = Q_j \tilde{S}_1(j) - Q'_j \tilde{S}_2(j) - O\&M_j$$
$$r^a = \text{discount rate (WACC, CAPM)}$$

Limitations:

- Difficult cash flow estimation
- Discount rate sensitivity
- Ignores volatility and optionality

Real Options Approach

$$V(0) = \sum_{j=1}^{N} C^j(0)$$

where C^j are spark spread options:

$$C^j(j) = \max(0, Q_1 S_1(j) - Q_2 S_2(j) - O\&M_j)$$

Advantages:

- Captures optionality
- No discount rate (Margrabe's formula)
- Accounts for price spikes

3.3.10 Conditions for Valid Real Options Approach

1. **Clear Option Structure:** Starting date, maturity T (may be infinite)
2. **Identified Risk Sources:** Underlying S or (S_1, S_2) clearly defined
3. **Liquid Markets:** Continuous trading for dynamic hedging
4. **Stochastic Process:** Appropriate models with available data
5. **Option Type Recognized:** European/American/compound impact pricing
6. **Market Completeness:** Unique valuation possible

Practical Warning

If conditions 3 and 4 fail (illiquid markets, poor process specification), real options approach becomes highly questionable.

Example: Peaker Plant Valuation

A gas-fired peaker's value depends on electricity price volatility. DCF ignores this, while real options capture the value of producing only during price spikes (above \$100/MWh).

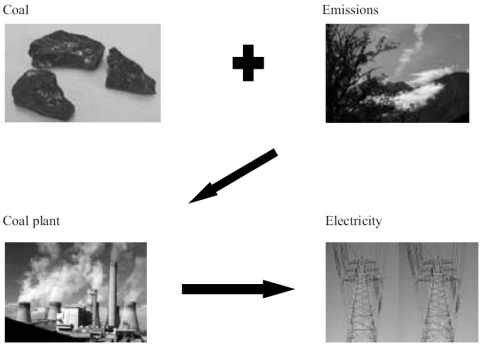
3.3.11 Dark Spread with Emissions: Kyoto Pricing of Coal Plants

For coal plants under emissions trading:

$$C(j) = \max(0, Q_1 S_1(j) - Q_2 S_2(j) - Q_3 E(j) - O\&M_j)$$

where:

- S_2: Coal price
- $E(j)$: Carbon price (per ton)
- Q_3: Tons of carbon emitted for Q_1 MWh



3.3.12 Commodity-Related Equities & M&A Activity

Alternative Exposure via Company Shares

- Traditional way to gain commodity exposure (oil/gold mining stocks)
- Particularly useful when futures don't exist for a commodity
- Example: Fertilizer companies during corn/wheat price spikes

Leveraged Exposure

- Commodity equities = leveraged way to invest in the commodity
- Gold mining stocks declined sharply with falling gold prices (2012-2014)
- Corporate governance and risk management affect earnings

M&A Activity in Agri-Food

Gigantic M&A activity in commodity trading houses and food companies (2012-2014):

- US companies competing with Nestlé
- China securing food supply (domestic & international)
- Creates merger arbitrage opportunities

3.3.13 Transportation as Locational Spread Options

Pipelines and LNG ships are options on **Locational Spreads**:

$$\text{Payoff} = \max(P_{dest} - P_{orig} - C_{transport}, 0) \tag{3.7}$$

Examples

- **LNG ship**: Option between Atlantic and Pacific basins
- **Pipeline**: Option to flow gas north-south or east-west
- **Oil tanker**: Option between regional markets

Key Valuation Factors

- Transportation time and costs
- Volatility of locational spreads
- Correlation between regional prices
- Option to idle or reposition empty

Other Spread Option Examples

- Flexible oil-indexed gas contract: Option on spread between oil and gas
- Dual-fired power plant: Option on spread between oil and gas
- Flexible transportation capacity: Option on locational spread between markets

3.3.14 Why Invest in Agricultural Land as Real Asset Investment?

Macro Drivers

- Population growth: 9.2B by 2050
- Biofuel demand: 240M acres needed
- Arable land loss: 0.5→0.2 ha/person
- Water scarcity: 90% China freshwater for irrigation
- Yield plateau: <1% annual increase since 2000

Investment Rationale

- Inflation hedge (real asset) diversification
- Food/energy scarcity positioning
- UK farmland: +134% in 10 years
- US corn-belt: +212% (2002-2012)

Global Land Price Comparison (USD/tonne corn)

- Central Illinois (USA): $1162
- Santa Fe (Argentina): $1065
- Matto Grosso (Brazil): $680
- Central Black Earth (Russia): $221
- Poltava (Ukraine): $113

3.3.15 Pricing Framework: The Black Model

To value options on futures (common for these assets), use Black model:

$$C_t = e^{-r(T-t)}[F_t N(d_1) - KN(d_2)] \tag{3.8}$$

$$d_1 = \frac{\ln(F_t/K) + \frac{\Sigma^2}{2}(T-t)}{\Sigma\sqrt{T-t}}$$

$$d_2 = d_1 - \Sigma\sqrt{T-t}$$

$$\Sigma = \sqrt{\frac{1}{T-t}\int_t^T \sigma^2(u, T')du}$$

Notations:

- F_t: Current Futures price
- K: Strike price
- r: Risk-free interest rate
- Σ: Average volatility over option life

Crucial Insight

For spread options, the **correlation** ρ between the two assets is the primary driver. As $\rho \uparrow$, spread volatility $\downarrow$, and the option value $\downarrow$.

3.3.16 Soybean Crusher / Ethanol Transformation Plant Valuation

Real Options Approach

Plant value = portfolio of spread options:

$$V(0) = \sum_{j=1}^{N} C^j(0)$$

$$C^j(j) = \max(0, Q_1 S_1(j) - Q_2 S_2(j) - O\&M_j)$$

Advantages:

- Captures optionality/opportunities
- No discount rate needed (Margrabe's formula)
- Accounts for spikes in output/low input prices

DCF Approach

$$V(0) = \sum_{j=1}^{n} \frac{E(\tilde{\phi}_j)}{(1+r^a)^j}$$

where $E(\tilde{\phi}_j) = Q_1 F_1^j(0) - Q_2 F_2^j(0) - O\&M_j$

Limitations:

- Fixed discount rate (r^a = WACC or CAPM)
- Locks sales/purchases at date 0
- Misses market optionalities

Practical Implementation

Acquirer typically: sells 80% output/input forward (intrinsic), keeps 20% optionality (extrinsic). Bid price = weighted average of both approaches.

3.3.17 Practical Challenges & Market Incompleteness

Warning! The risk-neutral valuation framework assumes:

1. Complete markets (all risks hedgeable)
2. Liquidly traded underlying
3. Continuous trading possible

These assumptions often fail for real options!

Sources of Incompleteness

- Volume/quantity risks
- Physical constraints
- Non-traded risks (political)
- Illiquid underlying markets
- Transaction costs

Alternative Approaches

- Utility-based pricing
- Mean-risk optimization
- Real-world measure + risk premium
- Intrinsic + extrinsic decomposition

Implied Volatility Surface

Market option prices imply volatility "smile": volatilities for high/low strikes > at-the-money. Reveals limitations of Black-Scholes (constant vol assumption fails).

Pedagogical Takeaways: Part III Synthesis

1. Strategy

Real Options: Engineering assets (plants/mines) must be valued by the manager's right to respond to prices.

Flexibility Premium: Swing/Storage are "convex" assets; they justify a price premium over standard swaps.

2. Assets

Agricultural: Land/Plants require specific Real Option analysis vs. traditional DCF models.

3. Risk Dynamics

Joint Risk: Managing spreads (Spark/Crack) requires modeling the *joint* correlation of inputs and outputs.

Volume Risk: $P \times Q = Revenue$. Negative correlation is a "Natural Hedge" but complicates pure price hedging.

4. Market Reality

Incomplete Markets: Unhedgeable risks mean "fair prices" are often ambiguous.

5. Numerics

Essential Toolkit:

- **Trees:** For American exercise.
- **Monte Carlo:** For high-dimensional assets.
- **LSMC:** To determine path-dependent continuation values.

Numerical convergence is the engineering standard for asset valuation.